Lecture L04
Information Theory

Judging forecasts and representations: the log score, entropy, KL divergence and mutual information

Probabilistic Machine Learning 2027-1

YOUR NOTES

Section 1

Forecasts, decisions and temperature

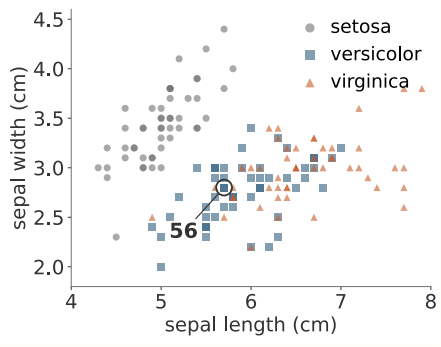
One fitted model serves several tasks: how does each task turn the forecast into its output?

YOUR NOTES

L04.1: Information theory (Forecasts and tasks)

A classifier fitted to the Iris flowers gives flower 56 class probabilities

Carried over (L01)
Iris: 150 labelled flowers, three species, four measurements each; $D = \{(x_n, y_n)\}$
A class model gives class probabilities and a label decision



The Iris data, sepal length and width.

The classification here

Data: $D = \{(x_n, y_n)\}_{n=1}^{150}$, $y \in \{\text{setosa}, \text{versicolor}, \text{virginica}\}$
Input: $x = (\text{sepal length}, \text{sepal width})$
Classifier: $\hat{y} = f_D(x)$
Class probabilities: $p(Y \mid x, D)$

Flower 56 (row 56, counted from 1)

$x_{56} = (5.7, 2.8)$, $y_{56} = \text{versicolor}$

	setosa	versicolor	virginica
$p(y \mid x_{56}, D)$	0.090	0.639	0.271

L01 S2.06 (14), S2.08 (16); Fisher (1936); scikit-learn `load_iris`; course-constructed: the classification

S1.02 · Slide 3/63

STUDY NOTE

The species overlap in the sepal plane, so the classifier returns a distribution over the three species instead of a single label, and flower 56 is a versicolor that still receives probability 0.271 for virginica. This running example supplies the forecast whose use and scoring the lecture develops.

YOUR NOTES

L04.1: Information theory (Forecasts and tasks)

The fit learns the forecast; the task chooses the decision and its measure

Carried over (L01)

Training
$f_T : E \to f_A$
E: experience

Task
$f_A : \text{In} \to \text{Out}$
f_A: learned function

Evaluation
$f_P : (\text{In}, \text{Out}) \to P$
P: performance
$\ell_{01}(y, \hat{y}) = \mathbb{I}\{y \neq \hat{y}\}$

Fit	Forecast	Decision	Performance
$D, p(Y \mid x, \theta), p(\theta)$ $\mapsto p(\theta \mid D)$	$q(Y \mid x) = p(Y \mid x, D)$ $= \int p(Y \mid x, \theta) p(\theta \mid D)\, \mathrm{d}\theta$	$f_A(x)$ $= \delta(q(Y \mid x))$	$P[f_A]$ $= \frac{1}{N} \sum_n \ell(y_n, f_A(x_n))$
the 150 flowers	*flower 56:* $q(Y \mid x_{56})$	*classification*	*flowers the fit did not use*
$D \mapsto \hat{\theta}$	$(0.090, 0.639, 0.271)$	$\hat{y} = \text{versicolor}$	$\ell = \ell_{01}$

$f_A = \delta \circ q$

q: learned by the fit

δ: chosen by the task

$q(Y \mid x)$: the distribution
$q(y \mid x)$: one probability
ℓ: chosen by the task

- A point fit $\hat{\theta}$ (maximum likelihood or MAP) is the posterior with all its probability on $\hat{\theta}$; the integral then becomes $p\left(Y \mid x, \hat{\theta}\right)$, the forecast with $\hat{\theta}$ plugged in.
- $P[f_A]$ is one number for the whole of f_A, on cases the fit did not use.

L01 S2.05 (13), S2.07 (15); S1.02; Book 1 §4.6.7.1, Eqs. 4.195–4.197

S1.03 · Slide 4/63

STUDY NOTE

The fit produces the forecast q(Y | x); the task then chooses the decision rule and the measure applied to it. Separating the two lets one fitted forecast serve different tasks, classification being the case where the rule picks a label and the measure is zero-one loss.

YOUR NOTES

L04.1: Information theory (Forecasts and tasks)

The best decision has the lowest expected cost under the forecast

Carried forward

forecast $q(Y \mid x) = \int p(Y \mid x, \theta) p(\theta \mid D)\, \mathrm{d}\theta$ decision $f_A(x) = \delta(q(Y \mid x))$

cost $P[f_A] = \frac{1}{N} \sum_n \ell(y_n, f_A(x_n))$

A second task on flower 56: virginica is poisonous, so eat it or not?

The eating costs, then flower 56′s forecast and expected costs

	setosa *edible*	versicolor *edible*	virginica *poisonous*	expected cost $\sum_y q(y \mid x_{56})\, \ell(y, a)$
$\ell(y, \text{eat})$	0	0	9	$0.090 \cdot 0 + 0.639 \cdot 0 + 0.271 \cdot 9 = 2.44$
$\ell(y, \text{refuse})$	1	1	0	$0.090 \cdot 1 + 0.639 \cdot 1 + 0.271 \cdot 0 = \mathbf{0.73}$ **→ refuse**
$q(y \mid x_{56})$	0.090	0.639	0.271	

- best decision $= \arg\min_a \sum_y q(y \mid x)\, \ell(y, a)$
- Eat only when $q(\text{edible} \mid x) > 0.9$.
- With zero-one cost the rule picks the most probable label, the **modal label**: the argmax rule.

S1.03; Book 1 §1.2.1.4–§1.2.1.5, Table 1.2; course-constructed: costs 9 and 1, threshold 0.9

STUDY NOTE

A decision is judged by weighting its cost over the outcomes by the forecast's probabilities, and the best decision minimizes that expected cost. With a large cost for eating a poisonous flower, flower 56 is refused although its most probable label is edible, while zero-one costs recover the argmax label.

YOUR NOTES

L04.1: Information theory (Forecasts and tasks)

One forecast serves several tasks, each with its own output and measure

Carried forward

flower 56 $q(y \mid x_{56}) = 0.090, 0.639, 0.271$ for y = setosa, versicolor, virginica

decision $f_A(x) = \delta(q(Y \mid x))$

eating costs $\ell(y, \text{eat}) = 0, 0, 9$; $\ell(y, \text{refuse}) = 1, 1, 0$

	Classify	Eat	Forecast
rule δ	$\hat{y} = \arg\max_y q(y \mid x)$	eat iff $q(\text{edible} \mid x) > 0.9$	$f_A(x) = q(Y \mid x)$
on flower 56: $f_A(x_{56})$	versicolor	**refuse**: $q(\text{edible} \mid x_{56}) = 0.729$	$(0.090, 0.639, 0.271)$
measure	ℓ_{01}	$\ell(y, a)$	?

- ℓ_{01} judges only a label; Eat needs its own costs.
- Forecast outputs the distribution itself: no measure for it yet.

S1.03–S1.04; Classify: L01 S2.08 (16); Forecast: Book 1 §5.1.6

S1.05 · Slide 6/63

STUDY NOTE

Classify, Eat and Forecast turn the same q into different outputs, so each task needs its own measure of performance. The first two have one, a zero-one loss and a cost table, which judge a label or an action; a task whose output is the distribution itself has none yet.

YOUR NOTES

L04.1: Information theory (Forecasts and tasks)

The classifier's forecast is a softmax of its logits

Carried over (L02)

categorical, θ: category probabilities	$p(c \mid \theta) = \theta_c, \theta_c > 0, \sum_c \theta_c = 1$
log ratios to a baseline K	$\eta_c = \ln(\theta_c/\theta_K), \eta_K = 0, \quad \theta_c = \theta_K e^{\eta_c}$
sum to one fixes θ_K: softmax	$\theta_c = e^{\eta_c} / \sum_k e^{\eta_k} = \text{softmax}(\eta)_c$

The classifier

logits $z_y(x)$, one for each label y, computed from the input x

$$q(y \mid x) = \text{softmax}(z(x))_y = \frac{e^{z_y(x)}}{\sum_k e^{z_k(x)}}$$

Here: $z_y(x) = w_y^T x + b_y$, linear in the sepals

- A classifier computes these coordinates from each input: the logits $z(x)$.
  - The categorical had one fixed η for every case.
- We take a fit of this form as given, with specific w_y and b_y; how to fit it is L05.

Flower 56

$z(x_{56}) = (-1.101, 0.860, 0)$

setosa, versicolor, virginica; virginica is the baseline

$e^{z_y(x_{56})} = 0.333,\ 2.363,\ 1;$ sum 3.696

$q(y \mid x_{56}) = e^{z_y(x_{56})}/3.696 = 0.090,\ 0.639,\ 0.271$

L02 S6.02–S6.07 (51–53); S1.02–S1.03; course computation: flower 56's logits

S1.06 · Slide 7/63

STUDY NOTE

The classifier has the softmax form of the earlier categorical family, but computes its coordinates, the logits, from each input instead of keeping one fixed set. Flower 56's probabilities come from exponentiating and normalizing its logits; the fit itself is taken as given.

YOUR NOTES

L04.1: Information theory (Forecasts and tasks)

Temperature tunes a forecast's uncertainty

Carried forward

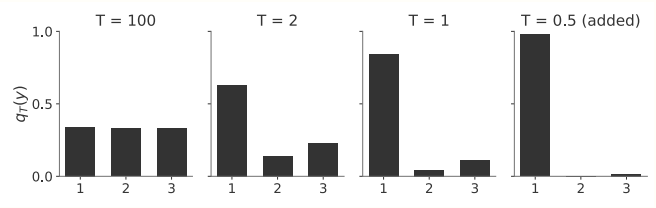
$q(y \mid x) = \text{softmax}(z(x))_y$ logits $z(x)$

Temperature

$q_T(y \mid x) \triangleq \text{softmax}(z(x)/T)_y, \quad T > 0$

$f_A = \delta \circ q \quad \rightarrow \quad f_A = \delta \circ q_T$

- Two reasons to tune T:
  - to repair a forecast that is too sure, or not sure enough, for the outcomes;
  - to bias the forecast on purpose toward a task.



$q_T(y) = \text{softmax}(z/T)_y$ for logits $z = (3, 0, 1)$

- $T \to \infty$: uniform.
- $T > 1$ flattens the forecast.
- $T = 1$: the fitted forecast.
- $T < 1$ sharpens it.
- $T \to 0$: all the probability on the modal label, the **point prediction**.

S1.03, S1.06; figure: redrawn from Book 2 §14.2.2.5, Fig. 14.3, p. 581 (= Book 1 Fig. 2.12), $T = 0.5$ added

S1.07 · Slide 8/63

STUDY NOTE

Dividing the logits by T turns one fit into a family of forecasts that differ only in how sharply they commit. Tuning T serves either to repair a forecast whose confidence does not match the outcomes or to bias it deliberately toward a task.

YOUR NOTES

L04.1: Information theory (Forecasts and tasks)

Temperature never changes the modal label

Two labels y, y', one input x, any $T > 0$

$z_y(x) > z_{y'}(x) \quad \Rightarrow \quad e^{z_y(x)/T} > e^{z_{y'}(x)/T}$

$q_T(y \mid x) > q_T(y' \mid x)$ both divided by the same sum

$\arg\max_y q_T(y \mid x) = \arg\max_y z_y(x) = \hat{y}$

- Classification's output, and so ℓ_{01}, is the same at every T.
- In the figure, $z = (3, 0, 1)$: label 1 has the largest logit, so the largest probability at every T.

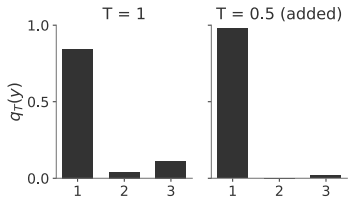
Carried forward

$$q_T(y \mid x) = \frac{e^{z_y(x)/T}}{\sum_k e^{z_k(x)/T}}$$

modal label, scored by ℓ_{01}

$\hat{y} = \arg\max_y q(y \mid x)$

Carried forward



S1.03, S1.06–S1.07; figure: S1.07's T = 1 and T = 0.5 panels

S1.08 · Slide 9/63

STUDY NOTE

Dividing every logit by the same positive T preserves their order, so the label with the largest logit stays the most probable at every T. Any measure that sees only the label therefore cannot register what temperature does to a forecast.

YOUR NOTES

L04.1: Information theory (Forecasts and tasks)

Temperature can change a decision that reads probabilities while the label stays

Carried forward

	$q(y \mid x_{56}) =$
flower 56	$0.090, 0.639, 0.271$
	setosa, versicolor, virginica
eating rule	eat iff $q(\text{edible} \mid x) > 0.9$
temperature	$q_T(y \mid x) \triangleq \text{softmax}(z(x)/T)_y$

Flower 56 at four temperatures

	$T = 1$	$T = 0.5$	$T = 0.25$	$T \to 0$
Modal label	versicolor	versicolor	versicolor	versicolor
$q_T(\text{edible} \mid x_{56})$	0.729	0.851	0.969	$\to 1$
Eat?	no	no	**yes**	**yes**

- ℓ_{01} cannot see the eating answer flip, since every label stays.
- The point prediction eats a flower with $q(\text{virginica} \mid x_{56}) = 0.271$.
  - Acting on the label alone is acting on the point prediction.

- A task that reads q changes its answer with T.
  - So a T is judged by that task's measure.

S1.03, S1.04, S1.07; course computation

S1.09 · Slide 10/63

STUDY NOTE

Flower 56's modal label stays versicolor at every temperature, yet the eating decision flips between T = 0.5 and T = 0.25 because it reads the probability of an edible flower. A temperature is therefore judged by the measure of a task that reads probabilities.

YOUR NOTES

L04.1: Information theory (Forecasts and tasks)

Section 1 summary: One forecast, several tasks; temperature keeps the labels

Task	Rule δ	Measure
Classify	$\hat{y} = \arg\max_y q(y \mid x)$	ℓ_{01}
Eat	eat iff $q(\text{edible} \mid x) > 0.9$	$\ell(y, a)$
Forecast	identity	S2

Flower 56	Modal label	$q_T(\text{edible} \mid x_{56})$	Eat?
$T = 1$	versicolor	0.729	no
$T = 0.5$	versicolor	0.851	no
$T = 0.25$	versicolor	0.969	yes
$T \to 0$	versicolor	$\to 1$	yes

TAKE HOME

- One forecast serves several tasks: each decision is the one with the lowest expected cost under the forecast, and each task has its own measure.
- Changing T changes the probabilities, never the labels, so ℓ_{01} cannot see it.
- A decision that reads the probabilities can change with T; a T is judged by that task's measure.

S1.05; flower 56: S1.09

S1.10 · Slide 11/63

STUDY NOTE

The section established that one forecast serves several tasks, each decision being the cost-minimizing one under that forecast, with its own measure. Temperature changes forecasts and the decisions that read them but never the labels, so zero-one loss is blind to it.

YOUR NOTES

Section 2

Scoring probability forecasts

Which forecast is better, and by what measure?

YOUR NOTES

L04.2: Information theory (Scoring forecasts)

The forecasting task outputs the forecast itself, not a label

Carried forward

$f_A(x) = \delta(q(Y \mid x))$ $\quad P[f_A] = \frac{1}{N}\sum_n \ell(y_n, f_A(x_n))$ $\quad \delta$ and ℓ chosen by the task

	Classify	Forecast
rule δ	$\hat{y} = \arg\max_y q(y \mid x)$	identity: $f_A(x) = q(Y \mid x)$
on flower 56: $f_A(x_{56})$	versicolor	$(0.090, 0.639, 0.271)$ setosa, versicolor, virginica
measure ℓ	ℓ_{01}	?

- ℓ_{01} compares labels: it cannot score a distribution.
- The forecasting task needs a measure that reads the probabilities.

S1.03, S1.05

S2.02 · Slide 13/63

STUDY NOTE

When a task's output is the forecast itself, its decision rule is the identity and what is needed is a measure that reads the probabilities. Zero-one loss compares labels, so it cannot score a distribution.

YOUR NOTES

L04.2: Information theory (Scoring forecasts)

The log score: improbable outcomes have higher cost

Carried forward
Forecasting task: $f_A(x) = q(Y \mid x)$ $\quad P[f_A] = \frac{1}{N}\sum_n \ell(y_n, f_A(x_n))$

Log score

- The forecast gives each label a probability $q(y \mid x)$.
  - The label that happens is y.

$\ell(y, f_A(x)) \triangleq -\ln q(y \mid x)$

Mean over N cases:
$P[f_A] = -\frac{1}{N}\sum_{n=1}^{N} \ln q(y_n \mid x_n)$

$q(y \mid x)$	$-\ln q(y \mid x)$	
0.999	0.001	Label strongly expected
0.99	0.010	
0.9	0.105	
0.5	0.693	
0.1	2.303	
0.01	4.605	
0.001	6.908	Label very surprising

- Lower is better.
  - Units are nats (natural logarithms).
- A confident forecast that is surprised costs a lot.
- The mean log score over cases is the negative log likelihood (NLL).
  - Independent cases: $-1/N$ times the log-likelihood, so maximum likelihood minimizes it.

S1.03, S2.02; L02 S4.03 (36); log loss: Book 1 §5.1.6.1, Eq. 5.43; NLL: Book 1 §1.2.1.6, Eqs. 1.13–1.15

S2.03 · Slide 14/63

STUDY NOTE

The log score charges a forecast the negative log of the probability it gave to what happened, so a confident forecast meeting a surprising outcome pays heavily. Its mean over cases is the negative log-likelihood, linking the score to maximum likelihood on the cases the fit used.

YOUR NOTES

L04.2: Information theory (Scoring forecasts)

Forecasts on new cases are judged by their true expected log score

Carried forward

Log score: $\ell(y, f_A(x)) = -\ln q(y \mid x)$

Expected cost of a decision a: $\sum_y q(y \mid x)\, \ell(y, a)$

weighted by the forecast q

Carried over (L01)

We want useful performance on future cases.

A new case at x: the expected log score

$Y \sim p^*(Y \mid x)$ p^*: the true distribution the outcomes come from; the star marks what is true

$E_{p^*}[-\ln q(Y \mid x)] = -\sum_y p^*(y \mid x) \ln q(y \mid x)$ weighted by p^*

- Each outcome counts as often as it happens, by p^*, not by the forecast's own q.
- So the expected log score judges q against p^*, and computing it needs p^*.
- That new cases come from a distribution is a hypothesis about the world.

S1.04, S2.03; L01 S2.07 (15); Book 1 §1.2.1.4

S2.04 · Slide 15/63

STUDY NOTE

The aim is performance on cases the fit has not seen, so the score is averaged over outcomes with the true distribution p* as weights, not the forecast's own q. The expected log score thus judges q against p*, and p* is a hypothesis about how new cases arise.

YOUR NOTES

L04.2: Information theory (Scoring forecasts)

A mean log score on held-out cases estimates the expected log score

Carried forward

Expected log score on a new case: $E[-\ln q(Y \mid X)]$, averaged over inputs too computing it needs p^*

The fit's own cases: it has seen them

Hold out M cases the fit did not use

Cases (x_m, y_m), drawn independently from the same $p^*(x, y)$

the hypothesis; q is fixed before they are scored; repeating the split is cross-validation

$\hat{S} \triangleq \frac{1}{M} \sum_{m=1}^{M} -\ln q(y_m \mid x_m)$

$E\left[\hat{S}\right] = \frac{1}{M} \sum_{m} E[-\ln q(Y_m \mid X_m)] = E[-\ln q(Y \mid X)]$ linearity; each case is a new case

$\mathrm{Var}\left(\hat{S}\right) = \sigma^2/M$, error $\sigma/\sqrt{M}$ σ: standard deviation of one case's score

- The held-out mean is right on average, for a fixed forecast.
- Its error shrinks as $1/\sqrt{M}$: with $\sigma = 0.671$, the spread of one case's score for the flowers of the next slide, it is 0.123 at $M = 30$, 0.095 at 50 and 0.055 at 150 nats.

S2.03, S2.04; course reasoning: the estimator and its standard error; course computation: the simulation's numbers

S2.05 · Slide 16/63

STUDY NOTE

Real data do not give p*, so the mean log score on held-out cases estimates the expected log score, without bias for a fixed forecast and with error shrinking like 1/sqrt(M). The fit's own cases cannot serve, because the fit has seen them.

YOUR NOTES

L04.2: Information theory (Scoring forecasts)

In simulation the true distribution is known and expected scores are computable

Carried forward

$q_T(y \mid x) \triangleq \text{softmax}(z(x)/T)_y$ $z(x)$: the classifier's logits

A simulation we control: the truth is the given fit, a function of the sepals; not a claim about real flowers

logits

$z_n = z(x_n)$

$z_y(x) = w_y^T x + b_y$

→ **true distribution**

$p_n^*(y) = \text{softmax}(z_n)_y$

→ **drawn once, kept**

$y_n \sim p_n^*$

→ **forecast at T**

$q_{n,T}(y) \triangleq q_T(y \mid x_n)$

$q_{n,1}(y) = p_n^*(y)$

What each makes computable

With p_n^*: $E_{p_n^*}\left[-\ln q_{n,T}(Y)\right] = -\sum_y p_n^*(y) \ln q_{n,T}(y)$ computable for every n and T

With y_n only: $-\ln q_{n,T}(y_n)$ all that real flowers give

- Real flowers give only y_n.
  - A held-out mean estimates the expected log score.

S1.02, S1.06–S1.07, S2.04; course-constructed: the simulation

S2.06 · Slide 17/63

STUDY NOTE

A simulation in which the fitted classifier is declared the truth makes both quantities available: expected scores computed exactly from p*, and the single-draw scores that real flowers would give. It is a laboratory for checking claims about scores, not a statement about real flowers.

YOUR NOTES

L04.2: Information theory (Scoring forecasts)

A sharper forecast can score better on most cases yet worse on average

Four simulated flowers at $T = 1$ and $T = 0.5$

n: x_n	y_n	$\hat{y}_n$ (both T)	$q_{n,1}$ (setosa, versicolor, virginica)	$-\ln q_{n,1}(y_n)$	$q_{n,0.5}$ (setosa, versicolor, virginica)	$-\ln q_{n,0.5}(y_n)$
61 (5.0, 2.0)	versicolor	versicolor	(0.056, **0.876**, 0.068)	0.13	(0.004, **0.990**, 0.006)	0.01
56 (5.7, 2.8)	versicolor	versicolor	(0.090, **0.639**, 0.271)	0.45	(0.017, **0.834**, 0.149)	0.18
141 (6.7, 3.1)	versicolor	virginica	(0.005, **0.294**, 0.701)	1.22	(0.000, **0.150**, 0.850)	1.90
131 (7.4, 2.8)	virginica	virginica	(0.000, 0.159, **0.840**)	0.17	(0.000, 0.035, **0.965**)	0.04

Mean log score

$T = 1$: $(0.13 + 0.45 + 1.22 + 0.17)/4 = 0.49$

$T = 0.5$: $(0.01 + 0.18 + 1.90 + 0.04)/4 = 0.53$

- Same labels, one mistake each (flower 141).
- $T = 0.5$ scores lower on the three flowers it gets right.
- Its mistake costs 1.90, not 1.22, so its mean is higher.

S2.03, S2.06; course computation

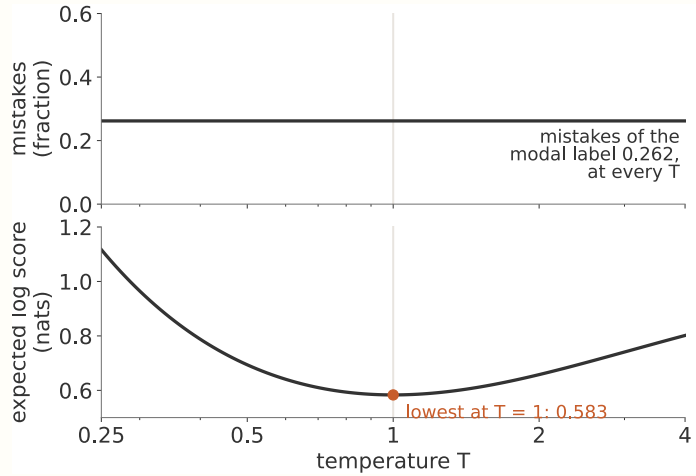
STUDY NOTE

At T = 0.5 the three correctly labelled flowers score better, but the one mistake costs enough more that the mean is worse, although both temperatures give the same labels. The log score weighs a confident mistake heavily, which zero-one loss, blind to confidence, cannot do.

YOUR NOTES

L04.2: Information theory (Scoring forecasts)

Mistakes stay flat; the expected log score is lowest at the true distribution



150 simulated flowers; expectations under the true distributions p^*_n; T on a log axis.

The two curves

mistakes of the modal label

$$\frac{1}{N}\sum_n \left(1 - \max_y p^*_n(y)\right)$$

expected log score

$$\frac{1}{N}\sum_n \sum_y p^*_n(y)\left(-\ln q_{n,T}(y)\right)$$

- No label moves, so every temperature makes the same mistakes.
  - Zero-one loss cannot tell the point prediction from the true distribution.
- The expected log score can: in this simulation it is lowest at $T = 1$, where $q_{n,1} = p^*_n$ by construction.

S1.07, S2.06; course computation

S2.08 · Slide 19/63

STUDY NOTE

Across temperatures the mistakes stay constant while the expected log score varies and is lowest at T = 1, where the forecast equals p* by construction. The log score thus separates forecasts that zero-one loss cannot, and ranks the true distribution best here.

YOUR NOTES

L04.2: Information theory (Scoring forecasts)

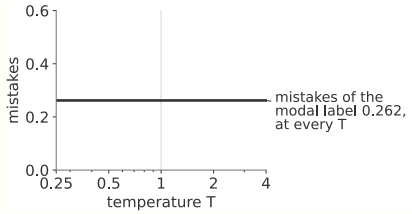
Section 2 summary: The log score scores the forecast; zero-one loss cannot

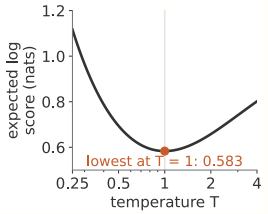
Forecasting task

$$f_A(x) = q(Y \mid x)$$

$$\ell(y, f_A(x)) = -\ln q(y \mid x)$$

$$-\frac{1}{N}\sum_{n=1}^{N} \ln q(y_n \mid x_n) \quad \text{mean: NLL}$$





TAKE HOME

- The log score, − ln of the probability given to what happened, scores the forecast itself; on its training cases the maximum-likelihood fit minimizes the mean.
- On new cases a forecast is judged by its expected log score under the true distribution; on the simulated flowers it was lowest there, while zero-one loss stayed flat.

S2.02–S2.03; figure: S2.08

S2.09 · Slide 20/63

STUDY NOTE

The section established the log score as the measure for a forecast, with its mean over training cases equal to the negative log-likelihood. On new cases a forecast is judged by its expected log score under the true distribution, which the simulation showed zero-one loss cannot track.

YOUR NOTES

Section 3

Entropy, cross-entropy and KL divergence

Is the expected log score always lowest at the outcomes' distribution? What are its floor and a forecast's excess?

YOUR NOTES

L04.3: Information theory (Entropy and KL)

Entropy measures how unpredictable an outcome is

Carried forward
Expected log score: $E_{p^*}[-\ln q(Y \mid x)] = -\sum_y p^*(y \mid x) \ln q(y \mid x)$

Entropy

$H(p) \triangleq E_p[-\ln p(Y)] = -\sum_y p(y) \ln p(y)$ for a distribution p over outcomes y; $0 \ln 0 = 0$

Five outcomes, entropy in nats

	$p(1)$	$p(2)$	$p(3)$	$p(4)$	$p(5)$	$H(p)$
certain	1	0	0	0	0	0
concentrated	0.9	0.025	0.025	0.025	0.025	0.464
spread	0.6	0.1	0.1	0.1	0.1	1.228
equally likely	0.2	0.2	0.2	0.2	0.2	1.609

- Log score: how surprising one outcome is. Entropy: how surprising p's outcomes are on average.
  - A measure of uncertainty: zero when one outcome is certain, growing as the probability spreads.
  - Here it is largest when all five are equally likely.

S2.03–S2.04; Book 1 §6.1, §6.1.1, Eq. 6.1; course-constructed: the table

STUDY NOTE

Entropy is the expected log score of a distribution evaluated under its own probabilities, the average surprise of its outcomes, and so it measures how unpredictable the outcome is. It is zero when one outcome is certain and largest when all outcomes are equally likely.

YOUR NOTES

L04.3: Information theory (Entropy and KL)

A forecast equal to p^* has expected log score $H(p^*)$

Carried forward

Expected log score: $E_{p^*}[-\ln q(Y \mid x)] = -\sum_y p^*(y \mid x) \ln q(y \mid x)$ Entropy: $H(p) = E_p[-\ln p(Y)]$

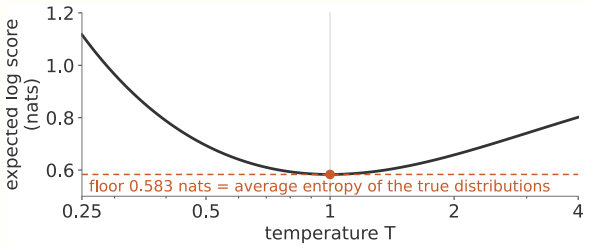
A forecast equal to p^*

One case, input fixed: write $p^*(y)$ for $p^*(y \mid x)$ and $q(y)$ for $q(y \mid x)$

$q = p^*$: $E_{p^*}[-\ln q(Y)] = E_{p^*}[-\ln p^*(Y)] = H(p^*)$

Simulated flowers, $T = 1$: $q_{n,1} = p^*_n$ by construction of the simulation; not true of real flowers

$\frac{1}{N}\sum_n E_{p^*_n}\left[-\ln q_{n,1}(Y)\right] = \frac{1}{N}\sum_n H(p^*_n) = 0.583$ nats



S2.08′s log-score panel, same axes

- 0.583 nats is how unpredictable the species are, even to a forecast that knows each p^*_n.
  - That forecast uses the sepals.

S2.04–S2.06, S2.08, S3.02; course computation

S3.03 · Slide 23/63

STUDY NOTE

Adding and subtracting ln p* inside the expectation splits the cross-entropy, the expected log score, into the entropy of p* and the KL divergence of q from p*: a floor no forecast goes below, plus the excess paid for differing from p*.

YOUR NOTES

L04.3: Information theory (Entropy and KL)

Any forecast's expected log score is an entropy plus a divergence

Split inside the expectation

$$H_{\text{ce}}(p^*, q) \triangleq E_{p^*}[-\ln q(Y)] \qquad \text{the expected log score, named: the cross-entropy}$$

$$= E_{p^*}\left[-\ln p^*(Y) + \ln \frac{p^*(Y)}{q(Y)}\right] \qquad \text{add and subtract } \ln p^*(Y)$$

$$= H(p^*) + E_{p^*}\left[\ln \frac{p^*(Y)}{q(Y)}\right] \qquad \text{the first term is the entropy}$$

The second term

$$\mathrm{KL}(p^* \parallel q) \triangleq E_{p^*}\left[\ln \frac{p^*(Y)}{q(Y)}\right] = \sum_y p^*(y) \ln \frac{p^*(y)}{q(y)}$$

$\sum_y$: over the possible outcomes y, not over cases

- The expected log score is the **cross-entropy** $H_{\text{ce}}(p^*, q)$.
- The second term is the **Kullback–Leibler (KL) divergence** of q from p^*.
- $H_{\text{ce}}(p^*, q) = H(p^*) + \mathrm{KL}(p^* \parallel q)$: the floor $H(p^*)$, plus the excess the forecast adds by differing from p^*.

S2.04, S3.02–S3.03; Book 1 §6.2.2, Eq. 6.31

S3.04 · Slide 24/63

STUDY NOTE

The fair-coin forecast of a coin with probabilities (0.8, 0.2) pays 0.193 nats above the floor of 0.500, a numerical instance of the split. KL is not symmetric: its first argument is the distribution the outcomes come from, and swapping gives a different number.

YOUR NOTES

L04.3: Information theory (Entropy and KL)

A fair-coin forecast of a biased coin pays 0.193 nats above the floor

Carried forward

$H_{\text{ce}}(p^*, q) = -\sum_y p^*(y) \ln q(y) = H(p^*) + \text{KL}(p^* \parallel q)$

$H(p^*) = -\sum_y p^*(y) \ln p^*(y), \;\; \text{KL}(p^* \parallel q) = \sum_y p^*(y) \ln \frac{p^*(y)}{q(y)}$

$p^* = (0.8, 0.2), \;\; q = (0.5, 0.5)$

$H_{\text{ce}} = -(0.8 \ln 0.5 + 0.2 \ln 0.5) = \ln 2 = 0.693$

$H(p^*) = -(0.8 \ln 0.8 + 0.2 \ln 0.2) = 0.500$ two outcomes: written $H_{\text{b}}(0.2)$, the binary entropy

$\text{KL}(p^* \parallel q) = 0.8 \ln \frac{0.8}{0.5} + 0.2 \ln \frac{0.2}{0.5} = 0.376 - 0.183 = 0.193$

$0.693 = 0.500 + 0.193$ nats

- The order matters: $\text{KL}(q \parallel p^*) = 0.5 \ln \frac{0.5}{0.8} + 0.5 \ln \frac{0.5}{0.2} = 0.223$, not 0.193.
  - The first argument is the distribution the outcomes come from.

S3.02, S3.04; course-constructed example

S3.05 · Slide 25/63

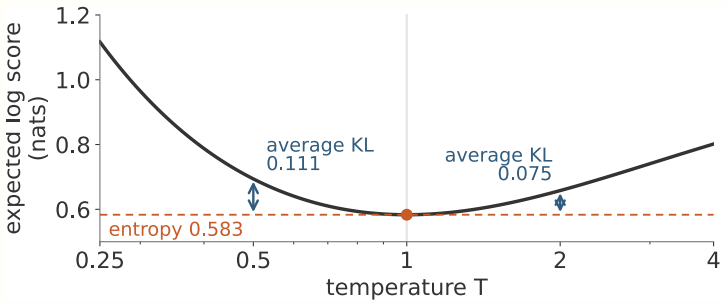
STUDY NOTE

The fair-coin forecast of a coin with probabilities (0.8, 0.2) pays 0.193 nats above the floor of 0.500, a numerical instance of the split. KL is not symmetric, so its first argument must be the distribution the outcomes come from, and swapping the arguments gives a different number.

YOUR NOTES

L04.3: Information theory (Entropy and KL)

On the flowers, the height above the floor is the forecasts' mean KL divergence



S2.08's log-score panel, same axes

Carried forward

Expected log score = cross-entropy:

$$H_{\text{ce}}(p^*, q) = H(p^*) + \text{KL}(p^* \parallel q)$$

Per flower, then averaged

$$\frac{1}{N}\sum_n H_{\text{ce}}(p_n^*, q_{n,T}) = \frac{1}{N}\sum_n H(p_n^*) + \frac{1}{N}\sum_n \text{KL}(p_n^* \parallel q_{n,T})$$

left side: the curve at T

- The mean KL is what the forecasts pay above the floor: 0.111 nats at $T = 0.5$, 0.075 at $T = 2$; computing it needs each p_n^*, so only the simulation can.
- $\text{KL}(p^* \parallel q) \geq 0$, zero only at $q = p^*$ (stated, from Murphy): for every p^* the expected log score is lowest only at $q = p^*$, which is what makes a score **strictly proper**.
  - Why it matters: a decision that reads the forecast, like eating, costs least with $q = p^*$.

S2.08, S3.03, S3.04; Book 1 Theorem 6.2.1 (proved there by Jensen's inequality); proper: Book 2 §14.2.1; course figure

S3.06 · Slide 26/63

STUDY NOTE

Averaging the split over the flowers shows the expected log score curve lies above its floor by the mean KL divergence of the forecasts from p*. Since KL is never negative and zero only at q = p*, the score is lowest only at the truth: it is strictly proper.

YOUR NOTES

L04.3: Information theory (Entropy and KL)

A zero probability on an outcome that happens gives an infinite log score

Carried forward

Log score: 6.908 nats at a probability of 0.001

Flower 141 at $T = 0.5$: $q_{141,0.5}(\text{versicolor}) = 0.150$, score 1.90 1.22 at $T = 1$

A zero probability on a possible outcome

If $p^*(y) > 0$ and $q(y) = 0$: $-\ln q(y) = \infty$

So $\mathrm{KL}(p^* \,\|\, q) = \infty$ and $H_{\text{ce}}(p^*, q) = \infty$

- The smaller the probability given to what happened, the larger the cost, without limit.
- A forecast that is never quite certain avoids an infinite score.

S2.03, S2.07, S3.04

S3.07 · Slide 27/63

STUDY NOTE

The cost of a forecast grows without bound as the probability it gives to what happens shrinks, and at exactly zero on a possible outcome the score, the cross-entropy and the KL divergence are all infinite. A forecast that keeps some probability on every possible outcome avoids this.

YOUR NOTES

L04.3: Information theory (Entropy and KL)

For continuous distributions, entropy becomes differential entropy

Carried forward

Finite outcomes: $H_{\text{ce}}(p,q) = H(p) + \mathrm{KL}(p \parallel q)$, $\mathrm{KL}(p \parallel q) \geq 0$, zero only at $q = p$

Densities $p(y)$, $q(y)$: sums become integrals

$$H_{\text{ce}}(p,q) = -\int p(y) \ln q(y)\, \mathrm{d}y, \quad \mathrm{KL}(p \parallel q) = \int p(y) \ln \frac{p(y)}{q(y)}\, \mathrm{d}y \geq 0$$

$H_{\text{ce}}(p,q) = h(p) + \mathrm{KL}(p \parallel q)$, with the **differential entropy** $h(p) \triangleq -\int p(y) \ln p(y)\, \mathrm{d}y$

Differential entropy is not an absolute amount of uncertainty

$Y \sim \mathrm{Uniform}(0, 1/8)$: $h = \ln(1/8) \approx -2.079$ nats negative: a density can exceed 1

Change units, $Z = aY$: $h(Z) = h(Y) + \ln|a|$ centimetres to millimetres adds $\ln 10 \approx 2.303$

- KL compares two densities, so the units cancel: it stays at least 0 and keeps its meaning as the excess.
- The floor $h(p)$ can be negative and depends on the units.

S3.02, S3.04, S3.06; Book 1 §6.1.6, Eqs. 6.24–6.25; §6.2.1, Eq. 6.30; course derivation: the change of units

S3.08 · Slide 28/63

STUDY NOTE

For densities the sums become integrals and the split keeps its form, with differential entropy h in place of H. Since h can be negative and depends on units while KL does not, the excess keeps its meaning but the floor is no absolute amount of uncertainty.

YOUR NOTES

L04.3: Information theory (Entropy and KL)

Quantized at bin width Δ, a continuous variable has entropy $\approx h(Y) - \ln \Delta$

Carried forward

Finite: $H(p) = -\sum_i p_i \ln p_i$ density: $h(p) = -\int p(y) \ln p(y)\, dy$

Bins of width Δ; Y_Δ: the bin Y falls in

$P(Y_\Delta = i) \approx p(y_i)\,\Delta$ y_i in bin i; Δ small

$$\begin{aligned} H(Y_\Delta) &\approx -\sum_i p(y_i)\Delta \ln(p(y_i)\Delta) \\ &= -\sum_i p(y_i)\Delta \ln p(y_i) - \ln\Delta \sum_i p(y_i)\Delta \quad \ln(ab) = \ln a + \ln b \\ &\approx -\int p(y) \ln p(y)\, dy - \ln\Delta \int p(y)\, dy \quad \text{sums} \to \text{integrals as } \Delta \to 0 \\ &= h(Y) - \ln\Delta \quad \int p(y)\, dy = 1 \end{aligned}$$

$N = 1/\Delta$ bins per unit of Y: $H(Y_\Delta) \approx h(Y) + \ln N$

Gaussian, $\sigma = 1$ cm

Δ	$H(Y_\Delta)$	$h - \ln\Delta$
1 cm	1.459	1.419
1 mm	3.722	3.722
0.1 mm	6.024	6.024

$h = \frac{1}{2}\ln(2\pi e) \approx 1.419$ nats

- Finer bins give more uncertainty, without limit: halving Δ adds $\ln 2$ nats, one bit, so a continuous value has infinite entropy.
- $h(Y)$ is the part that does not depend on the resolution; in the same units, a larger h means more spread, as for a Gaussian, $h = \frac{1}{2}\ln(2\pi e\sigma^2)$.

S3.02, S3.08; Book 1 §6.1.6 (quantization, citing Cover and Thomas); course computation: the Gaussian table

S3.09 · Slide 29/63

STUDY NOTE

Quantizing a continuous variable into narrow bins gives a discrete entropy of about h(Y) minus the log of the bin width, which grows without limit as the bins shrink. Differential entropy is the resolution-independent part, which is why differences such as KL carry over unchanged.

YOUR NOTES

L04.3: Information theory (Entropy and KL)

Section 3 summary: Entropy is the floor; KL is the excess

$$H_{\text{ce}}(p^*, q) = H(p^*) + \text{KL}(p^* \parallel q), \quad \text{e.g. } 0.693 = 0.500 + 0.193$$
$$\text{KL}(p^* \parallel q) \geq 0, \text{equality only at } q = p^*$$

TAKE HOME

- If the outcomes come from p^*, the entropy $H(p^*)$ is the lowest expected log score, that of a forecast equal to p^*.
- Any other forecast adds $\text{KL}(p^* \parallel q) \geq 0$, zero only at $q = p^*$ and infinite if it gives zero probability to what happens: the log score is strictly proper.
- With a strictly proper score, the true distribution is the forecast to rely on: on the flowers, the eating decision costs least at $T = 1$.

S3.04, S3.06

S3.10 · Slide 30/63

STUDY NOTE

The expected log score splits into the entropy of p*, the lowest score any forecast can have, and the KL divergence of the forecast from p*, which is never negative, zero only at p* and infinite for a zero probability on a possible outcome. The log score is strictly proper.

YOUR NOTES

Section 4

Maximum entropy and the exponential family

If experience gives us only averages, which distribution should we forecast with?

YOUR NOTES

L04.4: Information theory (Maximum entropy)

Entropy can choose a distribution when only feature averages are known

Carried over (L02)

Family chosen first: $p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

Moment identity: $\nabla A(\eta) = E_\eta[T(X)]$

Maximum likelihood: $E_{\hat{\eta}}[T(X)] = \frac{1}{N}\sum_{n=1}^{N} T(x_n)$ match the sample averages

- Fitting an exponential family chose the family first, then matched its averages to the data's.
- So ask the other way round: if we know only the averages, which distribution should we forecast with?

L02 S3.04 (27), S4.06 (39)

S4.02 · Slide 32/63

STUDY NOTE

Fitting an exponential family chose the family first and then matched its averages to those of the data. Here the order is reversed: when only averages are known, the question is which distribution to forecast with.

YOUR NOTES

L04.4: Information theory (Maximum entropy)

Choose the distribution that leaves the most uncertainty

Carried forward

Entropy: $H(p) = E_p[-\ln p(X)]$ outcome x; how surprising p's outcomes are on average

Outcomes $\{1, 2, 3\}$, mean 2.5

	p_1	p_2	p_3	rules out
end A	0	0.5	0.5	outcome 1
end B	0.25	0	0.75	outcome 2
a mixture	0.1	0.3	0.6	nothing

Looking for

The distribution with mean 2.5 and the most entropy

$$\max_{p:\, E_p[X]=2.5} H(p)$$

- Many distributions have mean 2.5; the average does not choose among them.
- The ends rule an outcome out, which the average gives no reason to do.
  - If it happens, the log score is infinite.
- Between the ends nothing is ruled out, and the distributions still differ.
  - Entropy measures how uncertain each leaves the outcome; the most uncertain assumes the least.

S3.02, S3.07; Book 2 §2.4.7; course-constructed example

S4.03 · Slide 33/63

STUDY NOTE

Knowing only that the mean is 2.5 leaves many distributions, and those ruling out an outcome risk an infinite log score. Entropy ranks the distributions by how much uncertainty each leaves, so choosing the one with the most entropy assumes as little as possible beyond the given average.

YOUR NOTES

L04.4: Information theory (Maximum entropy)

For mean 2.5, the maximum-entropy distribution is $(0.116, 0.268, 0.616)$

The sum and the mean leave one free number

$p_1 + p_2 + p_3 = 1, \;\; p_1 + 2p_2 + 3p_3 = 2.5$

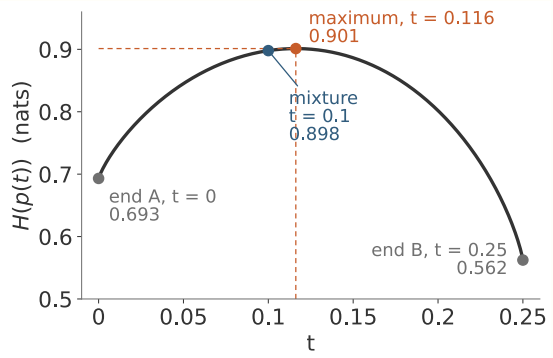
$t \triangleq p_1$: $\;\; p(t) = (t, 0.5 - 2t, 0.5 + t)$

$0 \le t \le 0.25$ from $p_1 \ge 0$ and $p_2 \ge 0$

The top of the curve

$t \approx 0.116$: $\;\; p \approx (0.116, 0.268, 0.616)$

$H \approx 0.901, \;\; p_2/p_1 = p_3/p_2 \approx 2.303$



Entropy along $p(t)$, in nats; the mixture is the table's $(0.1, 0.3, 0.6)$

- The maximum is the top of the curve, found numerically.
- It has no zero probability.
- Each probability is 2.303 times the one before it.

S4.03; course computation (generate_figures.py)

STUDY NOTE

The sum and the mean leave one free parameter, so the entropy becomes a curve in one variable and its maximum, found numerically, is (0.116, 0.268, 0.616). That maximum has no zero probability and equal ratios between successive probabilities, a pattern the next slide identifies.

YOUR NOTES

L04.4: Information theory (Maximum entropy)

On the example, the maximum-entropy distribution is an exponential family

Carried forward

The curve's top: $p \approx (0.116, 0.268, 0.616)$, each probability 2.303 times the one before

Carried over (L02)

$p_\eta(x) = h(x)\exp(\eta\,T(x) - A(\eta))$

Equal ratios, written as an exponential

Equal ratios $r = e^\lambda$: $p_x \propto r^x = e^{\lambda x}$ $\lambda = \ln 2.303 \approx 0.834$: the log of the ratio

Normalized: $p_\lambda(x) = \exp(\lambda x - A(\lambda))$, $A(\lambda) \triangleq \ln \sum_x e^{\lambda x}$: the family with $h = 1, T(x) = x, \eta = \lambda$

- Given only the mean, the maximum-entropy distribution is an exponential family whose statistic is x, the quantity whose average was given.

S4.04; L02 S1.02 (3); course computation

S4.05 · Slide 35/63

STUDY NOTE

Equal successive ratios mean the probabilities are proportional to an exponential of x, which is an exponential family with h = 1 and the statistic x, the quantity whose average was given. In this example the maximum-entropy distribution therefore belongs to the family used in fitting.

YOUR NOTES

L04.4: Information theory (Maximum entropy)

Maximum entropy with given feature averages is an exponential family

Carried forward
The example: given $E_p[X] = 2.5$ on $\{1, 2, 3\}$, $p_\lambda(x) = \exp(\lambda x - A(\lambda))$

	in the example	in general (derived in the recording)
outcomes	$x \in \{1, 2, 3\}$	$x \in \{1, \dots, L\}$
features	$f(x) = x$	$f(x) = (f_1(x), \dots, f_r(x))$, e.g. x and x^2
given averages	$\sum_x p_x\, x = 2.5$	$\sum_x p_x f(x) = F$, one F_j per feature
reference	$h(x) = 1$	$h(x)$, chosen: $h = 1$ for the most entropy; another h: closest in KL to it
result	$\exp(\lambda x - A(\lambda))$	$p_\lambda(x) = h(x) \exp(\lambda^T f(x) - A(\lambda))$, $A(\lambda) \triangleq \ln \sum_x h(x) e^{\lambda^T f(x)}$

- The family is fixed by the features and the choice of h; for the most entropy given only the averages, h is constant, the least restrictive choice.
- It is the only distribution with those averages and that much entropy.
- As a forecast it has the best guaranteed log score: it scores $H(p_\lambda)$ under every distribution with those averages, and no forecast has a better worst case (proved in the recording).
- With no features, p_λ is uniform and $H(p) \leq \ln L$.

S4.04–S4.05; L02 S1.02 (3); Book 2 §2.4.7, Eqs. 2.251–2.256 (Murphy's $-\lambda$ is our λ); derived in L04-R01

STUDY NOTE

For any features and given averages, the maximum-entropy distribution has the exponential-family form with those features as statistics, with its parameters set to match the averages. As a forecast it has the best guaranteed log score given them, a result derived in the recording.

YOUR NOTES

L04.4: Information theory (Maximum entropy)

On any outcome space, maximum entropy gives an exponential family

Carried forward
Finite outcomes, uniform reference: $p_\lambda(x) = \exp(\lambda^T f(x) - A(\lambda))$ differential entropy for densities

The general statement (stated)

Outcomes: x in a finite set, the integers, or $\mathbb{R}^d$

Reference (base measure): $h(x) > 0$ not the differential entropy $h(p)$; $h = 1$: uniform in the chosen coordinates

Given: the averages $E_p[f(X)] = F$ of features $f(x)$

Then: the p closest in KL to h is with $h = 1$, $\mathrm{KL}(p \parallel 1) = -h(p)$: the most entropy

$$p_\lambda(x) = h(x)\exp(\lambda^T f(x) - A(\lambda)), \quad A(\lambda) = \ln \int h(x) e^{\lambda^T f(x)}\, \mathrm{d}x$$ a sum for discrete x

- With no averages imposed, the answer is the reference h itself; on a finite set with $h = 1$, the uniform.
- Each imposed average tilts h by a factor $e^{\lambda_j f_j(x)}$; λ is set so the averages come out right, and h stays as chosen.
- On an infinite range a constant h is not a probability, and the maximum exists only if $he^{\lambda^T f}$ can be normalized: on $\mathbb{R}$, the mean alone gives none.

S3.08, S4.06; L02 S1.02 (3); Book 2 §2.4.7, Eqs. 2.251–2.256; for densities, not derived

S4.07 · Slide 37/63

STUDY NOTE

The same form holds on any outcome space, with the reference h as the distribution the answer stays closest to in KL; h = 1 gives the most entropy. Each imposed average tilts the reference, and on an infinite range a maximum exists only if the tilted reference normalizes.

YOUR NOTES

L04.4: Information theory (Maximum entropy)

With a given mean and variance, the maximum-entropy density is the Gaussian

Carried forward

$p_\lambda(x) = h(x)\exp(\lambda^T f(x) - A(\lambda)),\ \ A(\lambda) = \ln \int h(x) e^{\lambda^T f(x)}\,\mathrm{d}x$

$x \in \mathbb{R},\ \ h = 1,\ \ f(x) = (x, x^2),$ **mean μ, variance σ^2**

$p_\lambda(x) \propto \exp(\lambda_1 x + \lambda_2 x^2)$

$\lambda_1 = \mu/\sigma^2,\ \ \lambda_2 = -1/(2\sigma^2)$: $\ \lambda_1 x + \lambda_2 x^2 = -(x-\mu)^2/(2\sigma^2) + \text{const}$

so $\ p_\lambda = \mathcal{N}(\mu, \sigma^2)$ the constant goes into A

- $A(\lambda)$ is finite only for $\lambda_2 < 0$: the variance constraint is what makes the maximum exist.

S4.07; Book 2 §2.4.7 (end); course check of the parameters

S4.08 · Slide 38/63

STUDY NOTE

With features x and x squared, the given mean and variance make the exponent a quadratic, which is the Gaussian density, so the Gaussian is the maximum-entropy distribution for a given mean and variance. The variance constraint is what makes the normalizing integral finite.

YOUR NOTES

L04.4: Information theory (Maximum entropy)

Section 4 summary: Maximum entropy given averages is an exponential family

$$p_\lambda(x) = h(x)\exp(\lambda^T f(x) - A(\lambda))$$

Mean 2.5 on $\{1, 2, 3\}$: $p_\lambda \approx (0.116, 0.268, 0.616)$, $\lambda = \ln 2.303$

Mean and variance on $\mathbb{R}$: $\mathcal{N}(\mu, \sigma^2)$

TAKE HOME

- Among the distributions with given feature averages, the one with the most entropy is an exponential family in those features.
- On the mean-2.5 example, λ is the log of the equal ratio; with a mean and variance on $\mathbb{R}$ it is the Gaussian.
- In general its base measure $h(x)$ is the reference it stays closest to in KL.
- As a forecast it has the best guaranteed log score given those averages.

S4.03–S4.08

S4.09 · Slide 39/63

STUDY NOTE

Among distributions with given feature averages, the one with the most entropy is an exponential family in those features, with the base measure as the reference it stays closest to in KL. For a given mean and variance on the real line it is the Gaussian.

YOUR NOTES

Section 5

What an input is worth to a forecast

How much better can a forecast of the species be with the sepals than without them?

YOUR NOTES

L04.5: Information theory (What an input is worth)

Without the input, the best forecast is the label's marginal: here uniform

- The input X is a flower's sepals (length, width); the label Y is its species.

Carried forward

Simulated flowers: sepals $x_n \;\to\; p^*_n = p^*(Y \mid x_n) = \mathrm{softmax}(z(x_n))$, each flower with probability $1/150$

Expected log score: $H_{\mathrm{ce}}(p^*, q) = H(p^*) + \mathrm{KL}(p^* \parallel q)$, lowest, $H(p^*)$, at $q = p^*$

A forecast that ignores the input: the same $q(y)$ for every flower

True: over all flowers, species y occurs with $p^*_Y(y) = \frac{1}{150}\sum_n p^*_n(y) = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$ the label's marginal

Score of q: $\frac{1}{150}\sum_n H_{\mathrm{ce}}(p^*_n, q) = H_{\mathrm{ce}}(p^*_Y, q) = H(p^*_Y) + \mathrm{KL}(p^*_Y \parallel q)$ a cross-entropy

Best: $q = p^*_Y$, where the KL is 0, scoring the entropy $H(Y) \triangleq H(p^*_Y) = \ln 3 = 1.099$ nats

- Without the input, the best forecast is the species frequencies; they are equal here, so it is uniform.
- Its score, 1.099 nats, is the entropy of the label's marginal.

S1.06, S2.06, S3.04; course computation

S5.02 · Slide 41/63

STUDY NOTE

Without the input, the best forecast under the log score is the label's marginal distribution, and its score is the entropy of that marginal, here ln 3. This is the baseline against which the worth of the input is measured.

YOUR NOTES

L04.5: Information theory (What an input is worth)

With the input, the entropy must be averaged over the inputs

Carried forward

The truth, one fixed function of the sepals: $p^*(y \mid x) = \mathrm{softmax}(z(x))_y$, $z_y(x) = w_y^T x + b_y$ the given fit

At one input the best forecast is $p_n^* = p^*(Y \mid x_n)$, scoring $H(p_n^*)$ without the input: $H(Y) = 1.099$

From sepals to forecast to score, on four simulated flowers

sepals x_n (cm) (length, width); flower n	best forecast $p_n^* = \mathrm{softmax}(z(x_n))$ (setosa, versicolor, virginica)	its score $H(p_n^*)$
(5.0, 2.0); 61	(0.056, 0.876, 0.068)	0.461
(5.7, 2.8); 56	(0.090, 0.639, 0.271)	0.856
(6.7, 3.1); 141	(0.005, 0.294, 0.701)	0.635
(7.4, 2.8); 131	(0.000, 0.159, 0.840)	0.440

Averaged over the 150 flowers: the conditional entropy

$H(Y \mid X) \triangleq \frac{1}{150} \sum_n H(p_n^*) = 0.583$ nats

the entropy at each input, averaged over the inputs; the floor of the log-score curve

- The input is the sepals: each flower's sepals give its species distribution, its best forecast.
- Averaged over the flowers, the best score with the input is 0.583 nats; without it, 1.099.

S1.06, S2.06–S2.07, S3.03, S3.04, S5.02; Book 1 §6.1.4, Eqs. 6.11–6.15; course computation

S5.03 · Slide 42/63

STUDY NOTE

With the input, each flower has its own best forecast p*_n scoring H(p*_n), and averaging these over the inputs defines the conditional entropy H(Y | X). The same question, the best achievable score, has now been answered both without and with the input.

YOUR NOTES

L04.5: Information theory (What an input is worth)

Mutual information is how much knowing the input reduces the label's entropy

Carried forward

Best without the sepals: $H(Y) = 1.099$ best with the sepals: $H(Y \mid X) = 0.583$

Gain from the sepals: best score without, minus best score with

flower n: x_n	without: $H(Y)$	with: $H(p_n^*)$	gain
61: (5.0, 2.0)	1.099	0.461	0.638
56: (5.7, 2.8)	1.099	0.856	0.242
141: (6.7, 3.1)	1.099	0.635	0.464
131: (7.4, 2.8)	1.099	0.440	0.659

Averaged over the 150 flowers: mutual information

$$I(X;Y) \triangleq H(Y) - H(Y \mid X) = 1.0986 - 0.5833 = 0.5153 \approx 0.515 \text{ nats}$$

- No forecast that uses the sepals scores below 0.583 on average, and none that ignores them below 1.099.
- So 0.515 nats is the most the sepals can add to the best forecast without them: 47% of its 1.099.
- The gain varies by flower, from 0.040 to 1.042.

S3.04, S5.02, S5.03; course computation

S5.04 · Slide 43/63

STUDY NOTE

The difference between the best score without the input and the best score with it is the mutual information, the average amount by which knowing the input reduces the label's entropy. This answers how much better a forecast can be with the sepals than without them.

YOUR NOTES

L04.5: Information theory (What an input is worth)

Mutual information is the mean KL divergence of the conditional from the marginal

Carried forward

$H_{\text{ce}}(p^*, q) = H(p^*) + \text{KL}(p^* \parallel q)$ gain at an input: best score without it minus best score with it

Gains on four flowers: 0.638, 0.242, 0.464, 0.659

At input x_n: $Y \sim p_n^*$, forecast $q = p_Y^*$

$E_{p_n^*}[-\ln p_Y^*(Y)] = H(p_n^*) + \text{KL}(p_n^* \parallel p_Y^*)$ entropy plus KL

$E_{p_n^*}[-\ln p_Y^*(Y)] - H(p_n^*) = \text{KL}(p_n^* \parallel p_Y^*)$ gain at x_n: 0.638, 0.242, 0.464, 0.659 on the four flowers

$$I(X;Y) = \frac{1}{150}\sum_n \text{KL}(p_n^* \parallel p_Y^*) = 0.515 \text{ nats}$$

- At one input, the gain is how far the label's distribution there is from its marginal, measured by KL.
- Mutual information averages it over the inputs: the expected KL from the marginal (before seeing the input) to the conditional (after).
- Every gain is at least 0, since KL is.

S3.04, S5.04; L01 S2.08 (16); Book 2 §5.1.2 ("information gain"); course reading

S5.05 · Slide 44/63

STUDY NOTE

Applying the entropy-plus-KL split to the marginal forecast at each input shows that the gain there is the KL divergence from the marginal to the conditional distribution. Mutual information averages these gains, a divergence from before the input is seen to after, so it is never negative.

YOUR NOTES

L04.5: Information theory (What an input is worth)

Mutual information is symmetric and zero exactly under independence

Carried forward

$I(X;Y) = H(Y) - H(Y \mid X) = 0.515$ nats $\mathrm{KL}(p \parallel q) \geq 0$, equality only at $p = q$

From the difference of entropies to one KL divergence; $(X, Y) \sim p^*$

$$
\begin{aligned}
I(X;Y) &= H(Y) - H(Y \mid X) && \text{the best gain} \\
&= E[-\ln p_Y^*(Y)] - E[-\ln p^*(Y \mid X)] && \text{each entropy as an expectation} \\
&= E\left[\ln \frac{p^*(Y \mid X)}{p_Y^*(Y)}\right] = E\left[\ln \frac{p^*(X,Y)}{p_X^*(X)\, p_Y^*(Y)}\right] && \text{one expectation; times } p_X^*(X)/p_X^*(X) \\
&= \mathrm{KL}(p_{XY}^* \parallel p_X^*\, p_Y^*) && p_X^* p_Y^*\text{: independent, same margins}
\end{aligned}
$$

- X and Y enter alike: $I(X;Y) = I(Y;X)$; the species tell as much about the sepals as the sepals about the species.
- $I \geq 0$, and $I = 0$ exactly when the joint is the product: the input is worth nothing exactly when the label is independent of it.

S3.06, S5.03–S5.04; Book 2 §5.3.1–§5.3.2, Eqs. 5.109–5.111

S5.06 · Slide 45/63

STUDY NOTE

Writing the difference of entropies as one expectation shows mutual information is the KL divergence of the joint distribution from the product of its margins. It treats the variables alike, so I(X; Y) = I(Y; X), and is zero exactly when label and input are independent.

YOUR NOTES

L04.5: Information theory (What an input is worth)

Replacing p^* with fitted forecasts changes what the gain measures

Carried forward

$I(X;Y) = 0.515$ nats needs p^* a held-out mean estimates an expected log score

What real flowers allow: fit, then measure on held-out cases

Fit a classifier $q(Y \mid x)$ (the given form, on the training flowers) and a baseline $q_0(Y)$ that ignores the sepals

gain $\triangleq E[-\ln q_0(Y)] - E[-\ln q(Y \mid X)]$ estimated by the difference of two held-out means

In the simulation, where p^* is known: a classifier fitted to 30 flowers

baseline q_0	gain (expected)	against $I = 0.515$
the 30 flowers' class frequencies, $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	0.443	below
a poor baseline, $(0.8, 0.1, 0.1)$	0.953	above

- With p^*, the gain is $I(X;Y)$: a property of the input and the label alone.
- With fitted forecasts, it also depends on the classifier and the baseline: 0.443 and 0.953 against 0.515.

S1.06, S2.05, S5.04; course computation on the simulation

S5.07 · Slide 46/63

STUDY NOTE

With real flowers only fitted forecasts exist, so a measured gain is the difference between the held-out scores of a baseline and a classifier. That gain also depends on the particular classifier and baseline, and it can fall below or above I(X; Y), as the two baselines show.

YOUR NOTES

L04.5: Information theory (What an input is worth)

A fitted gain is mutual information corrected by two divergences

Carried forward

One case: $E_{p^*}[-\ln q(Y)] = H(p^*) + \mathrm{KL}(p^* \parallel q)$ $\quad I(X;Y) = H(Y) - H(Y \mid X)$

Split each expected score, then subtract

$E[-\ln q_0(Y)] = H(Y) + \mathrm{KL}(p^*_Y \parallel q_0)$ the split, with $p^* = p^*_Y$

$E[-\ln q(Y \mid X)] = H(Y \mid X) + E_X \mathrm{KL}(p^*(Y \mid X) \parallel q(Y \mid X))$ the split at each input, averaged

$\text{gain} = [H(Y) + \mathrm{KL}(p^*_Y \parallel q_0)] - [H(Y \mid X) + E_X \mathrm{KL}(p^*(Y \mid X) \parallel q(Y \mid X))]$ subtract the two lines

$= \underbrace{H(Y) - H(Y \mid X)}_{I(X;Y)} + \mathrm{KL}(p^*_Y \parallel q_0) - E_X \mathrm{KL}(p^*(Y \mid X) \parallel q(Y \mid X))$ regroup

On the flowers (each term rounded to three decimals)

Class frequencies: $0.443 \approx 0.515 + 0 - 0.073$

Poor baseline: $0.953 \approx 0.515 + 0.511 - 0.073$

- A poor baseline inflates the gain; a forecast short of $p^*(Y \mid X)$ shrinks it.
- So a measured gain is not an estimate of mutual information.

S3.04, S5.04, S5.07; course computation on the simulation

S5.08 · Slide 47/63

STUDY NOTE

A fitted gain equals the mutual information plus the baseline's divergence from the true marginal minus the classifier's average divergence from the true conditional. A poor baseline inflates the gain and an imperfect classifier shrinks it, so a measured gain does not estimate mutual information.

YOUR NOTES

L04.5: Information theory (What an input is worth)

Section 5 summary: The sepals' worth is the difference of two best scores

$$H(Y) = \ln 3 = 1.099 \quad \text{without the sepals,} \quad H(Y \mid X) = 0.583 \quad \text{with them}$$

$$I(X;Y) = H(Y) - H(Y \mid X) = \frac{1}{150} \sum_n \mathrm{KL}(p^*_n \parallel p^*_Y) = \mathrm{KL}(p^*_{XY} \parallel p^*_X p^*_Y) = 0.515 \text{ nats}$$

TAKE HOME

- Without the sepals the best forecast is the species frequencies; with them, each flower's own.
- Their difference, mutual information, is the best gain in expected log score from the sepals: 0.515 nats.
- It is the expected KL divergence from the species frequencies to a flower's distribution, and the joint's KL divergence from independence.
- A fitted forecast's gain is not mutual information.

S5.02–S5.08

S5.09 · Slide 48/63

STUDY NOTE

The sepals' worth is the difference of two best scores, the marginal's entropy without them and the conditional entropy with them. This mutual information is also an expected KL divergence and the joint's KL divergence from independence; a gain measured with fitted forecasts differs from it.

YOUR NOTES

Section 6

What processing an input cannot do

Can a summary of the input tell us more about the label than the input itself?

YOUR NOTES

L04.6: Information theory (Processing an input)

A summary can keep all of the input's information about the label, some, or none

Carried forward
$I(X;Y) = H(Y) - H(Y \mid X) = 0.515$ simulated flowers: p^*_n a function of the sepals, each flower $1/150$

A summary $Z = g(X)$ is computed from the sepals alone
$I(Y;Z) = H(Y) - H(Y \mid Z)$, as for X

Summary $Z = g(x)$, $x = (x_1, x_2)$ = (length, width)	Values of Z	$I(Y;Z)$
$10x$ millimetres, not centimetres	117	0.515
x_1 length alone	35	0.423
$(x_1 + x_2)/2$ the average	38	0.214
$\arg\max_c p^*(c \mid x)$	3	0.456
1	1	0

- No summary tells more about the species than the sepals: none exceeds 0.515.
- A recoding keeps all of it and a constant none.
- Length alone keeps 0.423 of the 0.515; the average of length and width keeps only 0.214.

S5.04, S5.06; course computation on the simulation, not real flowers

S6.02 · Slide 50/63

STUDY NOTE

The question is whether a summary computed from the sepals can say more about the species than the sepals themselves. The table shows summaries that keep all, some or none of the 0.515 nats, and none keeps more, which the following slides establish in general.

YOUR NOTES

L04.6: Information theory (Processing an input)

Five definitions describe what a summary keeps about the label

Carried forward
$I(Y;X) = H(Y) - H(Y \mid X)$ $H(Y \mid X)$: the label's entropy at each input, averaged over inputs

Summary	$Z = g(X)$	computed from the input alone
Markov chain $Y \to X \to Z$	$p^*(z \mid x, y) = p^*(z \mid x)$	Z depends on Y only through X
Information from a pair	$I(Y;X,Z) = H(Y) - H(Y \mid X,Z)$	the pair (X,Z) as one input
Conditional mutual information	$I(Y;Z \mid X) = H(Y \mid X) - H(Y \mid X,Z)$	what Z adds about Y once X is known
Sufficient summary	$I(Y;Z) = I(Y;X)$	Z keeps all of X's information about Y

How to think about it

A sufficient summary is a compression that loses nothing about this label: we could keep Z and discard X. In the table, the sepals in millimetres are sufficient; sepal length alone is not.

S5.03–S5.04, S6.02; L03 S3.05; Book 1 §6.3.4; Book 2 §5.3.3–§5.3.4

S6.03 · Slide 51/63

STUDY NOTE

The definitions give the vocabulary: a Markov chain makes the summary depend on the label only through the input, pair information and conditional mutual information measure what a second variable adds, and sufficiency means keeping all the input's information.

YOUR NOTES

L04.6: Information theory (Processing an input)

Why processing cannot add information: the route

The route through this section

1. **The claim: for a Markov chain $Y \to X \to Z$, $I(Y;Z) \leq I(Y;X)$**
2. Conditional mutual information is never negative: $I(Y;Z \mid X) \geq 0$; under the chain it is 0
3. Information from a pair adds up: $I(Y;X,Z) = I(Y;X) + I(Y;Z \mid X)$, and with X and Z swapped
4. Put together: $I(Y;Z) \leq I(Y;X)$, with gap $I(Y;X \mid Z)$
5. Sufficient exactly when X adds nothing once Z is known: $I(Y;X \mid Z) = 0$

S6.05, S6.07, S6.09, S6.11, S6.13

STUDY NOTE

The proof of the inequality uses five facts, kept in view as a route and returned to before each is shown, beginning with the claim to be proved.

YOUR NOTES

L04.6: Information theory (Processing an input)

Processing an input cannot raise its information about the label

Carried over (L03)



A pipe: influence flows through the middle until the middle is known. As a definition: $p^*(z \mid x, y) = p^*(z \mid x)$; every summary $Z = g(X)$ gives one.

Data-processing inequality

For a Markov chain $Y \to X \to Z$: $I(Y;Z) \leq I(Y;X)$

- In the table, every summary is at most 0.515.
- No forecast that sees only Z can beat, in expected log score, the best one that sees X; the loss is $I(Y;X) - I(Y;Z)$.
- For any other task too, the best expected cost from Z is no lower than from X.
- A forecast fitted to Z can still beat one fitted to X: the divergence terms of a fitted gain depend on the fit.

S5.04, S6.02; L03 S3.03, S3.05; Book 2 §5.3.3, Theorem 5.3.1 (letters renamed); course reasoning: the order for every task

STUDY NOTE

Any summary computed from the input forms a Markov chain with the label, and the inequality says its information about the label cannot exceed the input's. No forecast seeing only the summary beats the best seeing the input; fitted forecasts are not covered.

YOUR NOTES

L04.6: Information theory (Processing an input)

Why processing cannot add information: the route

The route through this section

1. The claim: for a Markov chain $Y \to X \to Z$, $I(Y;Z) \leq I(Y;X)$
2. **Conditional mutual information is never negative: $I(Y;Z \mid X) \geq 0$; under the chain it is 0**
3. Information from a pair adds up: $I(Y;X,Z) = I(Y;X) + I(Y;Z \mid X)$, and with X and Z swapped
4. Put together: $I(Y;Z) \leq I(Y;X)$, with gap $I(Y;X \mid Z)$
5. Sufficient exactly when X adds nothing once Z is known: $I(Y;X \mid Z) = 0$

S6.05, S6.07, S6.09, S6.11, S6.13

S6.06 · Slide 54/63

STUDY NOTE

With the claim stated, the next fact needed is that conditional mutual information cannot be negative, and that it vanishes under the chain.

YOUR NOTES

L04.6: Information theory (Processing an input)

Conditional mutual information is never negative

Carried forward

$I(Y;Z \mid X) = H(Y \mid X) - H(Y \mid X,Z)$ KL ≥ 0, zero only between equal distributions

Chain: $p^*(z \mid x,y) = p^*(z \mid x)$, so $p^*(y,z \mid x) = p^*(y \mid x)\,p^*(z \mid x,y) = p^*(y \mid x)\,p^*(z \mid x)$

Conditional mutual information as a KL

$$I(Y;Z \mid X) = E\left[\ln \frac{p^*(Y \mid X,Z)}{p^*(Y \mid X)}\right] \quad \text{each entropy as an expectation}$$

$$= \sum_x p_X^*(x) \sum_{y,z} p^*(y,z \mid x) \ln \frac{p^*(y,z \mid x)}{p^*(y \mid x)\,p^*(z \mid x)} \quad p^*(y \mid x,z) = p^*(y,z \mid x)/p^*(z \mid x)$$

$$= \sum_x p_X^*(x)\,\mathrm{KL}(p^*(Y,Z \mid x) \parallel p^*(Y \mid x)\,p^*(Z \mid x)) \quad \text{the inner sum is a KL}$$

Each KL is ≥ 0 (zero iff $p^*(y,z \mid x) = p^*(y \mid x)\,p^*(z \mid x)$ for all y,z)

$I(Y;Z \mid X) \geq 0$ (zero iff Y and Z are independent given X)

Markov chain $Y \to X \to Z$: $I(Y;Z \mid X) = 0$

- Within each slice $X = x$, the mutual information between Y and Z; then averaged over x.

S3.06, S5.06, S6.05; L03 S3.05; Book 1 §6.3.4, Eqs. 6.56–6.57 (letters permuted)

S6.07 · Slide 55/63

STUDY NOTE

Conditional mutual information is a KL divergence averaged over the inputs, so it is never negative, and it is zero when label and summary are independent given the input, as under the chain. These are the two facts about it that the proof needs.

YOUR NOTES

L04.6: Information theory (Processing an input)

Why processing cannot add information: the route

The route through this section

1. The claim: for a Markov chain $Y \to X \to Z$, $I(Y;Z) \leq I(Y;X)$
2. Conditional mutual information is never negative: $I(Y;Z \mid X) \geq 0$; under the chain it is 0
3. **Information from a pair adds up: $I(Y;X,Z) = I(Y;X) + I(Y;Z \mid X)$, and with X and Z swapped**
4. Put together: $I(Y;Z) \leq I(Y;X)$, with gap $I(Y;X \mid Z)$
5. Sufficient exactly when X adds nothing once Z is known: $I(Y;X \mid Z) = 0$

STUDY NOTE

The next fact is that the information from a pair of variables splits additively, in either order.

YOUR NOTES

L04.6: Information theory (Processing an input)

Information from a pair of variables adds up, in either order

Carried forward

$I(Y;X) = H(Y) - H(Y \mid X)$ $I(Y;X,Z) = H(Y) - H(Y \mid X,Z)$ $I(Y;Z \mid X) = H(Y \mid X) - H(Y \mid X,Z)$

Where we will end: $I(Y;X,Z) = I(Y;X) + I(Y;Z \mid X)$

$H(Y)$: uncertainty about Y with no input

$I(Y;X)$: removed by X	$I(Y;Z \mid X)$: then by Z	$H(Y \mid X,Z)$: what remains

$I(Y;X,Z)$: removed by the pair

Add and subtract $H(Y \mid X)$

$$I(Y;X,Z) = H(Y) - H(Y \mid X,Z) = \underbrace{H(Y) - H(Y \mid X)}_{I(Y;X)} + \underbrace{H(Y \mid X) - H(Y \mid X,Z)}_{I(Y;Z \mid X)}$$

Chain rule

$I(Y;X,Z) = I(Y;X) + I(Y;Z \mid X) = I(Y;Z) + I(Y;X \mid Z)$ the second: Z first, the same steps

- What the pair tells about Y is what X tells, plus what Z adds once X is known.

S5.03, S5.04, S6.07; Book 1 §6.3.4, Eqs. 6.61–6.62; Book 2 Eqs. 5.114–5.115

S6.09 · Slide 57/63

STUDY NOTE

Adding and subtracting H(Y | X) splits the pair's information into what the input tells plus what the summary adds once the input is known; exchanging the variables gives a second expansion. The proof equates the two.

YOUR NOTES

L04.6: Information theory (Processing an input)

Why processing cannot add information: the route

The route through this section

1. The claim: for a Markov chain $Y \to X \to Z$, $I(Y;Z) \leq I(Y;X)$
2. Conditional mutual information is never negative: $I(Y;Z \mid X) \geq 0$; under the chain it is 0
3. Information from a pair adds up: $I(Y;X,Z) = I(Y;X) + I(Y;Z \mid X)$, and with X and Z swapped
4. **Put together: $I(Y;Z) \leq I(Y;X)$, with gap $I(Y;X \mid Z)$**
5. Sufficient exactly when X adds nothing once Z is known: $I(Y;X \mid Z) = 0$

STUDY NOTE

With the earlier facts in hand, the inequality follows by combining them.

YOUR NOTES

L04.6: Information theory (Processing an input)

Under the Markov chain, $I(Y;Z) \leq I(Y;X)$

Carried forward

$I(Y;X,Z) = I(Y;X) + I(Y;Z \mid X) \quad I(Y;X,Z) = I(Y;Z) + I(Y;X \mid Z)$

Chain: $I(Y;Z \mid X) = 0 \quad I(Y;X \mid Z) \geq 0$

For the Markov chain $Y \to X \to Z$

Equate the two expansions of $I(Y;X,Z)$:

$I(Y;Z) + I(Y;X \mid Z) = I(Y;X) + I(Y;Z \mid X)$

With $I(Y;Z \mid X) = 0$: $I(Y;Z) + I(Y;X \mid Z) = I(Y;X)$

$I(Y;X \mid Z) \geq 0 \quad \Rightarrow \quad I(Y;Z) \leq I(Y;X)$

- The gap $I(Y;X) - I(Y;Z)$ is exactly $I(Y;X \mid Z)$: what X still tells about Y once Z is known.
- With Z the sepal length: $0.515 - 0.423 = 0.092$ nats, what the width adds once the length is known.

S6.02–S6.09; Book 2 §5.3.3, Eq. 5.116

S6.11 · Slide 59/63

STUDY NOTE

Equating the two expansions, dropping the term that is zero under the chain, and using non-negativity of the remaining conditional mutual information gives the inequality, with gap I(Y; X | Z). For the sepal length the gap is what the width adds.

YOUR NOTES

L04.6: Information theory (Processing an input)

Why processing cannot add information: the route

The route through this section

1. The claim: for a Markov chain $Y \to X \to Z$, $I(Y;Z) \leq I(Y;X)$
2. Conditional mutual information is never negative: $I(Y;Z \mid X) \geq 0$; under the chain it is 0
3. Information from a pair adds up: $I(Y;X,Z) = I(Y;X) + I(Y;Z \mid X)$, and with X and Z swapped
4. Put together: $I(Y;Z) \leq I(Y;X)$, with gap $I(Y;X \mid Z)$
5. **Sufficient exactly when X adds nothing once Z is known: $I(Y;X \mid Z) = 0$**

S6.05, S6.07, S6.09, S6.11, S6.13

STUDY NOTE

The last step reads the gap in the inequality as a criterion for sufficiency.

YOUR NOTES

L04.6: Information theory (Processing an input)

A summary is sufficient exactly when the input adds nothing once it is known

Carried forward

Sufficient: $I(Y;Z) = I(Y;X)$ gap: $I(Y;X) - I(Y;Z) = I(Y;X \mid Z)$

$I(Y;X \mid Z) \geq 0$ (zero iff Y and X are independent given Z)

Sepals in millimetres: $I(Y;Z) = I(Y;X) = 0.515$ length alone: gap 0.092

> Z is sufficient iff $I(Y;X \mid Z) = 0$ iff Y and X are independent given Z

- The sepals in millimetres are sufficient: Z determines X, so X adds nothing given Z.
- Sepal length alone is not: the gap is positive.
- The same definition with a parameter θ in place of the label: $s(X)$ is sufficient for θ when $I(\theta; s(X)) = I(\theta; X)$.
  - For exponential families, the statistic $\sum_n T(x_n)$ of N observations is sufficient for the parameter.

S6.02, S6.07, S6.11; L02 S4.04 (37); Book 2 §5.3.4, Eq. 5.117

STUDY NOTE

A summary is sufficient exactly when the input adds nothing once it is known. The sepals in millimetres meet this and length alone does not; with a parameter in place of the label, the same definition gives the sufficient statistic of the exponential family.

YOUR NOTES

L04.6: Information theory (Processing an input)

Section 6 summary: Processing cannot add information about the label

$$Y \to X \to Z \quad \Rightarrow \quad I(Y;Z) \le I(Y;X)$$

$$I(Y;Z) + I(Y;X \mid Z) = I(Y;X), \quad \text{the sepals:} \quad 0.515 \ge 0.423 \ \text{(length alone)}$$

TAKE HOME

- A summary computed from the sepals, $Z = g(X)$, tells no more about the species than the sepals do: no processing of the input can raise the best achievable score, $I(Y;Z) \le I(Y;X)$.
- The proof: mutual information of a pair adds up in either order, the Markov chain removes one term, and conditional mutual information is never negative.
- The gap is $I(Y;X \mid Z)$, what the sepals still tell once Z is known; it is zero exactly when Z is sufficient, as for a recoding.

S6.02–S6.13

S6.14 · Slide 62/63

STUDY NOTE

No summary of the input tells more about the label than the input does, by the chain rule for pair information, the term vanishing under the chain, and non-negativity of conditional mutual information. The gap is what the input still adds, zero exactly for a sufficient summary.

YOUR NOTES

L04.NB: Information theory (Notebook)

Notebook: score Iris forecasts by accuracy and log score

In the notebook

1. Held-out accuracy and mean log score of prepared Iris forecasts, against the training class frequencies q_0
2. A same-labels, different-confidence comparison
3. A reliability summary, and what it can and cannot show

Carried forward

1. The mean log score, infinite at a zero probability on what happens; the held-out mean estimates the expected one; the best baseline p^*_Y; fitted gains are not mutual information
2. Same labels, different log scores; strict propriety
3. Calibration: the predicted label's accuracy against its confidence, bin by bin

S2.03, S2.05, S2.06, S2.07–S2.08, S3.06, S3.07, S5.02, S5.07–S5.08

NB.01 · Slide 63/63

STUDY NOTE

The notebook applies the lecture's quantities to prepared Iris forecasts: held-out accuracy and mean log score against class frequencies, forecasts with the same labels but different confidence, and a reliability summary of confidence against accuracy, a check the notebook introduces.

YOUR NOTES